

Science communication tools: rubrics for generating posters and manuscripts that are authentic to the practice of science

Tamarah Lynne Adair,¹ Swati Agrawal,² Yesmi P. Ahumada-Santos,³ Regina Alvarez,⁴ Kirk R. Anders,⁵ Mauricio Antunes,⁶ Mary Ayuk,⁷ María Elena Báez-Flores,³ Dondra Bailey,⁸ Frederick N. Baliraine,⁹ Mitchell F. Balish,¹⁰ Christa Tobey Bancroft,¹¹ Michèle Barmoy,¹² Tonya C. Bates,¹³ Andrea Beyer,¹⁴ Suparna Bhalla,¹⁵ Dave Bollivar,¹⁶ Lisa M. Bono,¹⁷ Rebecca Bortz,¹⁸ Tejas Bouklas,¹⁹ Kristen Butela,¹⁸ Christine A. Byrum,²⁰ Megan Carroll,²¹ Steven Michael Caruso,²² Nancy Castro,¹¹ Kari Clase,²³ Kristen Clermont,²⁴ Sean T. Coleman,²⁵ Pamela L. Connerly,²⁶ Amaya M. Garcia Costas,²⁷ Steven Gaines Cresawn,²⁸ Tom D'Elia,²¹ Aimee Danley,¹⁸ Megan K. Dennis,²⁹ Arturo Diaz,³⁰ Erin L. Doyle,³¹ Iain Duffy,³² Dustin Cole Edwards,³³ Elvira Eivazova,³⁴ Eric Engstrom,³⁵ Christy Fillman,³⁶ Christine L. Fleischacker,³⁷ Victoria J. Frost,³⁸ Julie Torruellas Garcia,³⁹ Bryan P. Gibb,⁴⁰ Alyssa Mae Gleichsner,⁴¹ Sharon Gusky,⁴² Rebekah F. Hare,⁴³ Danielle Heller,⁴⁴ Lee E. Hughes,⁶ Jacob D. Kagey,⁴⁵ Bridgette Kirkpatrick,⁴⁶ Karen K. Klyczek,⁴⁷ Kathryn P. Kohl,³⁸ Hari Kotturi,⁴⁸ Julia Y. Lee-Soety,⁴⁹ Heather Lindberg,⁵⁰ Bhaswati Manish,⁵¹ Sergei A. Markov,⁵² Matthew David Mastropaolo,⁵³ James T. Melton,⁵⁴ Jon Mitchell,⁵⁵ Sally Dixon Molloy,⁵⁶ Denise L. Monti,⁴⁴ Tiara Perez Morales,⁵⁷ María Alejandra Mussi,⁵⁸ Holly Nance,⁵⁹ Fernando Nieto,⁶⁰ Imade Yolanda Nsa,⁶¹ Shallee T. Page,⁶² Ricardo Parra-Unda,³ Vipaporn Phuntumart,⁶³ Brett E. Pickett,⁶⁴ Richard Scott Pollenz,⁶⁵ Sarah N. Reardon,⁶⁶ Jessica Rocheleau,⁶⁷ Ombeline Rossier,⁶⁸ Naomi Smith Rowland,⁶⁹ Adam D. Rudner,⁷⁰ Elizabeth E. Rueschhoff,²⁶ Viknesh Sivanathan,⁴⁴ Martha Smith Caldas,⁷¹ C. Nicole Sunnen,⁴⁹ Sarah J. Swerdlow,⁷² Kissaou Tchedre,⁷³ Kara Thoenke,⁷⁴ Deborah Tobiason,⁷⁵ Sara S. Tolsma,⁷⁶ Tara Turley-Stoulig,⁷⁷ Quinn Vega,⁷⁸ Patricia Waikel,⁷⁹ Catherine Ward,⁸⁰ Vassie C. Ware,⁸¹ Jackie M. Washington,⁸² Daniel Eric Westholm,⁷⁴ Beth Wilkes,⁸³ Daniel Williams,⁸⁴ Elizabeth Williams,⁷⁰ Ellen M. Wisner¹³

AUTHOR AFFILIATIONS See affiliation list on p. 4.

ABSTRACT Traditional undergraduate science courses often prioritize content mastery over authentic engagement with the scientific process. Course-based research, also referred to as course-based undergraduate research experiences (CUREs), addresses this limitation by immersing students in authentic scientific practice. In course-based research, assessment practices can also mirror the authentic scientific practice, where extensive formative feedback supports refinement of skills and understanding. Here, we present two rubrics designed to support the teaching and assessment of science communication in a way that reflects how scientists prepare to disseminate their research findings. One rubric is for creating scientific posters and another for writing short-format manuscripts. Developed by approximately 100 faculty members who collaboratively implement CUREs through the Howard Hughes Medical Institute (HHMI) Science Education Alliance (SEA) program, these rubrics outline the authentic steps scientists take when preparing to communicate their research and provide performance levels that clarify expectations for both students and instructors. Together, these tools aim to further align undergraduate science education and authentic scientific practice.

KEYWORDS course-based undergraduate research experiences (CUREs), science communication, assessment rubrics, authentic scientific practice, scientific posters, short-format manuscripts

Traditional undergraduate science courses frequently focus on the transmission of content, coupled with assessments focused on the recitation of facts and demonstration of laboratory skills. Such educational experiences have been criticized for failing to capture the authentic practice of science and their limited impact on student engagement and persistence in STEM fields (1, 2). By contrast, course-based research aims to engage students in the practices of professional science, where students leverage

Editor Melinda Tsaoying Owens, University of California San Diego, San Diego, California, USA

Address correspondence to Viknesh Sivanathan, sivanathanv@hhmi.org.

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science content and skills to investigate questions with unknown outcomes, deal with failure, iterate their work, collaborate, and disseminate their findings (1–3). As modeled by the Howard Hughes Medical Institute (HHMI) Science Education Alliance (SEA) program, which coordinates two course-based research projects (SEA-PHAGES and SEA-GENES) implemented at over 150 colleges and universities nationwide, assessment in these courses also aligns closely with the practice of professional science (4). In SEA course-based research projects, formal assessment early in the course focuses on establishing foundational knowledge, skills, and expectations that prepare students for subsequent research tasks. This is complemented by extensive *ad hoc*, informal formative feedback throughout the research process, which guides student learning and supports iterative improvement. Summative assessment at the end of the course evaluates students' mastery of core concepts, scientific reasoning, and research practices (4).

Hanauer et al. (4) describe four primary assessment aims in course-based research: (i) assessing laboratory work and scientific thinking, (ii) evaluating mastery of concepts, quantitative thinking, and skills, (iii) appraising forms of scientific communication, and (iv) promoting metacognition of learning (4). A survey of the literature suggests that effective scaffolding of science communication in course-based research remains underdeveloped (5, 6). To address this gap, faculty within the HHMI SEA program developed resources to authentically model how scientists prepare to communicate their science, in particular in preparing a scientific poster and a short-format manuscript for publication. Over 100 SEA faculty participated in a collaborative, iterative process to identify authentic scientific communication practices and translate them into performance-based rubrics refined through consensus. The deliverables are rubrics that are designed to teach students how scientists engage in science communication and to support meaningful assessment of student learning. By using these rubrics, students should be able to develop, refine, and communicate a coherent scientific message supported by appropriate evidence, using practices that mirror how scientists prepare posters and manuscripts for dissemination.

PROCEDURE

Using the rubrics for teaching and assessment

Both the poster and manuscript rubrics (Appendix 1 and 2) are designed to guide students through the authentic steps of preparing for scientific communication, from organizing ideas and data to presenting findings with clarity and accuracy. The rubrics include performance levels (Aware, Emerging, and Proficient) for each step, reflecting increasing levels of competence in scientific communication. For example, in the *Develop a Key Message* criterion of the poster rubric, students are tasked with distilling the central finding of their research into a concise statement understandable within five to ten seconds. At the aware level, the message is unclear, overly long, or poorly supported by the data, and scientific terminology may be inaccurate. At the emerging level, the key message is present and generally supported by the data but may be wordy or lack precision. At the proficient level, the key message is clear, concise, accurate, and easily understood by the intended audience; it invites conversation and is well supported by the presented data. When provided to students, the rubric serves as a transparent guide that helps both students and faculty align their expectations.

While obtaining feedback is listed as a step in the rubrics, the rubrics are designed to facilitate cycles of ongoing feedback and revision; instructors should not shy away from providing iterative feedback that models for students the rigor and precision expected in scientific communication, where constructive critique is given in service of both the individual researcher and the advancement of science. For formative assessment, the rubrics are most effective when used to provide targeted, constructive feedback. Instructors should provide review and feedback on individual components of the work at a time, such as background or key message, using the relevant rubric criteria. Feedback should highlight strengths and provide concrete, actionable suggestions tied to rubric

descriptors. Both instructors and peers can provide formative feedback. For summative assessment, instructors can use the performance levels to holistically assess the overall quality of the work, recognizing both mastery and growth. Summative evaluation should focus on how well students have incorporated feedback, improved clarity and accuracy, and demonstrated scientific thinking. Indeed, the goal of ongoing formative feedback is to support all students to reach proficiency. By combining formative use of the rubric for iterative improvement with summative use for final evaluation, instructors can create a coherent assessment approach.

Implementation tips and resources for preparing a poster

The rubric references several additional resources to enrich the teaching and learning experience. A first set of resources introduces students to principles of strong scientific communication using examples of well-designed posters. We recommend introducing students to the #BetterPoster framework developed by Mike Morrison (osf.io/ef53g), which emphasizes clear, minimalist design and rapid communication of key findings. We also recommend teaching students accessible and inclusive design principles for selecting color palettes, using adequate contrast between text and background, and choosing readable font types and sizes. To support this, instructors can leverage widely available guidance on inclusive design for scientific poster preparation, for example, as suggested by the American Chemical Society (<https://www.acs.org/content/dam/acsorg/about/inclusion/accessible-science-communication-pdf-ready.pdf>).

Once posters are drafted, a checklist (Appendix 3) is provided to help students to systematically review each element of their poster—verifying that the key message is clear, the figures and tables are appropriately labeled and support the key message, and accessibility standards are met—before printing or presenting.

Finally, to also help students leverage the poster session as an opportunity for idea exchange and professional growth, consider providing students with prompts for when they engage with a poster presenter. Some examples of guiding prompts are as follows. (i) What was the key message of the poster? (ii) How could the information in the poster inform your own research? (iii) Was there something from your own research that could help advance the presenter's work? (iv) What poster design features might you take away after visiting this poster?

Implementation tips and resources for writing a short-format manuscript

Course-based research projects are often focused on well-defined research outputs that contribute meaningfully to the scientific community, even when the scope of the research is relatively narrow. Within the SEA-PHAGES program, for instance, students isolate novel phages from environmental samples and manually annotate sequenced phage genomes. Such discrete research outcomes are suited for concise manuscript formats such as a *Genome Announcement*. The rubric described here was designed to guide and assess the preparation of this particular type of manuscript, although its structure can readily be adapted to other short-form research publications.

The rubric references several resources to facilitate manuscript preparation and good research practices. Chief among these is the Data Collection Template (Appendix 4), which helps students compile and organize all relevant experimental information before writing begins. This template should be introduced early in the research process so that students can record critical information, such as software versions and databases accessed, at the time of use. Because such tools and resources evolve rapidly, documenting them as they are used ensures that the research remains reproducible and that the original data are complete, traceable, and readily available should there be a need to revisit the raw data after the students have finished the course.

A second resource, the Manuscript Template (Appendix 5), provides section-by-section guidance on what content belongs in each part of the manuscript. The template features examples and cues about discriminating between necessary and extraneous

information, how to structure paragraphs, and how to maintain clarity and brevity consistent with short-format publications.

A final resource is the Checklist (Appendix 6), which assists instructors and students in verifying that all essential components are accurately included and formatted prior to submission.

CONCLUSION

These rubrics were developed by SEA faculty and reflect the instructional and assessment practices in SEA course-based research, where assessments of student outcomes have demonstrated significant gains in science identity, self-efficacy, and project ownership, all of which are linked to increased persistence in STEM (3, 4, 7). Moreover, existing research suggests that integrating structured communication tasks into course-based research enhances students' confidence, communication skills, and ability to think like scientists (5). It is worth noting that these rubrics have been evaluated implicitly through authentic scientific dissemination. For example, the manuscript rubric has been used to guide students in developing nearly 100 genome announcement manuscripts that have successfully undergone external peer review and been published in scientific journals. This outcome is not intended to serve as causal measures of student learning; rather, it provides externally validated evidence that the communication practices scaffolded by the rubrics meet the professional standards of the scientific community.

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AUTHOR AFFILIATIONS

¹Biology, Baylor University, Waco, Texas, USA

²Biology, University of Mary Washington College of Arts and Sciences, Fredericksburg, Virginia, USA

³Unidad de Investigaciones en Salud Pública Dra. Kaethe Willms, Facultad de Ciencias Químico Biológicas, Universidad Autónoma de Sinaloa, Culiacán, Sinaloa, Mexico

⁴Dominican College of Blauvelt, Orangeburg, New York, USA

⁵Biology, Gonzaga University, Spokane, Washington, USA

⁶University of North Texas, Denton, Texas, USA

⁷Biology, Howard University, Washington, DC, USA

⁸Coppin State University, Baltimore, Maryland, USA

⁹Biology & Kinesiology, LeTourneau University, Longview, Texas, USA

¹⁰Department of Microbiology, Miami University, Oxford, Ohio, USA

¹¹University of Southern California, Culver City, California, USA

¹²Allegany College of Maryland, Cumberland, Maryland, USA

¹³Department of Biological Sciences, UNC Charlotte Department of Biological Sciences, Charlotte, North Carolina, USA

¹⁴Richard Bland College, South Prince George, Virginia, USA

¹⁵Mount Saint Mary College, Newburgh, New York, USA

¹⁶Indiana University Bloomington, Bloomington, Indiana, USA

¹⁷Department of Biological Sciences, Texas Tech University, Lubbock, Texas, USA

¹⁸Department of Biological Sciences, University of Pittsburgh, Pittsburgh, Pennsylvania, USA

¹⁹University of California Los Angeles, Los Angeles, California, USA

- ²⁰Biology, College of Charleston, Charleston, South Carolina, USA
- ²¹Department of Biology, Indian River State College, Fort Pierce, Florida, USA
- ²²Department of Biological Sciences, University of Maryland Baltimore County, Baltimore, Maryland, USA
- ²³Purdue University, West Lafayette, Indiana, USA
- ²⁴La Salle University, Philadelphia, Pennsylvania, USA
- ²⁵Biology, Wartburg College, Waverly, Iowa, USA
- ²⁶Biology, Indiana University Southeast, New Albany, Indiana, USA
- ²⁷Biology, Colorado State University Pueblo, Pueblo, Colorado, USA
- ²⁸Biology, James Madison University, Harrisonburg, Virginia, USA
- ²⁹Marist College, Poughkeepsie, New York, USA
- ³⁰Biology, La Sierra University, Riverside, California, USA
- ³¹Biology Department, Doane University, Crete, Nebraska, USA
- ³²Saint Leo University, Saint Leo, Florida, USA
- ³³Biological Sciences, Tarleton State University, Stephenville, Texas, USA
- ³⁴Columbia State Community College, Columbia, Tennessee, USA
- ³⁵Biology, Monmouth College, Monmouth, Illinois, USA
- ³⁶University of Colorado Boulder, Boulder, Colorado, USA
- ³⁷Department of Biology, University of Mary, Bismarck, North Dakota, USA
- ³⁸Biology, Winthrop University, Rock Hill, South Carolina, USA
- ³⁹Biological Sciences, Nova Southeastern University, Fort Lauderdale, Florida, USA
- ⁴⁰Biological and Chemical Sciences, New York Institute of Technology, Old Westbury, New York, USA
- ⁴¹Biological Sciences, SUNY Plattsburgh, Plattsburgh, New York, USA
- ⁴²Connecticut State Community College – Northwestern Campus, Winsted, Connecticut, USA
- ⁴³Biology, Gonzaga University, Spokane, Washington, USA
- ⁴⁴Howard Hughes Medical Institute, Chevy Chase, Maryland, USA
- ⁴⁵Biology, University of Detroit Mercy, Detroit, Michigan, USA
- ⁴⁶Collin College, Plano, Texas, USA
- ⁴⁷University of Wisconsin-River Falls, River Falls, Wisconsin, USA
- ⁴⁸West Virginia School of Osteopathic Medicine, Lewisburg, West Virginia, USA
- ⁴⁹Biology, Saint Joseph's University, Philadelphia, Pennsylvania, USA
- ⁵⁰Virginia Western Community College, Roanoke, Virginia, USA
- ⁵¹Metropolitan Community College, Omaha, Nebraska, USA
- ⁵²Biology, Austin Peay State University, Clarksville, Tennessee, USA
- ⁵³Sciences, Neumann University, Aston, Pennsylvania, USA
- ⁵⁴Spelman College, Atlanta, Georgia, USA
- ⁵⁵Northern State University, Aberdeen, South Dakota, USA
- ⁵⁶Molecular and Biomedical Sciences, The University of Maine, Orono, Maine, USA
- ⁵⁷Department of Biological, Physical, and Health Sciences, Roosevelt University College of Arts and Sciences, Chicago, Illinois, USA
- ⁵⁸Centro de Estudios Fotosintéticos y Bioquímicos, Rosario, Santa Fe, Argentina
- ⁵⁹Natural Sciences, College of Coastal Georgia, Brunswick, Georgia, USA
- ⁶⁰SUNY Old Westbury, Old Westbury, New York, USA
- ⁶¹Microbiology, University of Lagos, Lagos, Nigeria
- ⁶²College of Health and Natural Sciences, Franklin Pierce University, Rindge, New Hampshire, USA
- ⁶³Bowling Green State University, Bowling Green, Ohio, USA
- ⁶⁴Microbiology and Molecular Biology, Brigham Young University, Provo, Utah, USA
- ⁶⁵Molecular Biosciences, University of South Florida, Tampa, Florida, USA
- ⁶⁶Biology, Culver-Stockton, Canton, Missouri, USA
- ⁶⁷Biology, University of Massachusetts Amherst, Amherst, Massachusetts, USA
- ⁶⁸Université Paris-Saclay, Gif-sur-Yvette, France
- ⁶⁹Western Kentucky University, Bowling Green, Kentucky, USA

⁷⁰University of Ottawa, Ottawa, Canada

⁷¹Division of Biology, Kansas State University, Manhattan, Kansas, USA

⁷²Biological Sciences, University of Pittsburgh Greensburg, Greensburg, Pennsylvania, USA

⁷³Austin Community College District – Eastview Campus, Austin, Texas, USA

⁷⁴The College of St Scholastica, Duluth, Minnesota, USA

⁷⁵Carthage College, Kenosha, Wisconsin, USA

⁷⁶Biology, Northwestern College of Iowa, Orange City, Iowa, USA

⁷⁷Southeastern Louisiana University, Hammond, Louisiana, USA

⁷⁸Montclair State University, Montclair, New Jersey, USA

⁷⁹Florida International University, Miami, Florida, USA

⁸⁰Durham Technical Community College, Durham, North Carolina, USA

⁸¹Biological Sciences, Lehigh University, Bethlehem, Pennsylvania, USA

⁸²Empire State University, Saratoga Springs, New York, USA

⁸³Natural Sciences, NHTI-Concord's Community College, Concord, New Hampshire, USA

⁸⁴Coastal Carolina University, Conway, South Carolina, USA

AUTHOR ORCIDS

Tamarah Lynne Adair  <http://orcid.org/0000-0002-4656-6769>

Kirk R. Anders  <http://orcid.org/0000-0002-3372-1379>

María Elena Báez-Flores  <http://orcid.org/0000-0003-1389-3568>

Frederick N. Baliraine  <http://orcid.org/0000-0002-4054-2467>

Mitchell F. Balish  <http://orcid.org/0000-0002-2606-5481>

Christa Tobey Bancroft  <http://orcid.org/0000-0002-7359-954X>

Lisa M. Bono  <http://orcid.org/0000-0001-5404-0946>

Kristen Butela  <http://orcid.org/0000-0003-1463-5288>

Christine A. Byrum  <http://orcid.org/0000-0002-6653-7922>

Steven Michael Caruso  <http://orcid.org/0000-0001-9166-9137>

Kari Clase  <http://orcid.org/0000-0003-2749-6440>

Sean T. Coleman  <http://orcid.org/0000-0003-3609-1402>

Amaya M. Garcia Costas  <http://orcid.org/0000-0002-8500-5489>

Steven Gaines Cresawn  <http://orcid.org/0000-0002-2162-562X>

Tom D'Elia  <http://orcid.org/0000-0003-2601-3128>

Arturo Diaz  <http://orcid.org/0000-0002-4174-8950>

Dustin Cole Edwards  <http://orcid.org/0000-0001-6409-8041>

Victoria J. Frost  <http://orcid.org/0000-0003-4929-0873>

Bryan P. Gibb  <http://orcid.org/0000-0002-2094-344X>

Lee E. Hughes  <http://orcid.org/0000-0001-6357-2696>

Jacob D. Kagey  <http://orcid.org/0000-0002-6994-1795>

Karen K. Klyczek  <http://orcid.org/0000-0001-7882-0429>

Julia Y. Lee-Soety  <http://orcid.org/0000-0002-9667-3175>

Sergei A. Markov  <http://orcid.org/0000-0002-4520-4539>

Matthew David Mastropaolo  <http://orcid.org/0000-0002-8148-4742>

Sally Dixon Molloy  <http://orcid.org/0000-0002-0478-1209>

María Alejandra Mussi  <http://orcid.org/0000-0002-4168-3624>

Imade Yolanda Nsa  <http://orcid.org/0000-0001-5606-268X>

Shallee T. Page  <http://orcid.org/0000-0002-3557-4122>

Ricardo Parra-Unda  <http://orcid.org/0000-0001-5291-6879>

Brett E. Pickett  <http://orcid.org/0000-0001-7930-8160>

Richard Scott Pollenz  <http://orcid.org/0000-0001-8891-6053>

Jessica Rocheleau  <http://orcid.org/0000-0003-1088-7277>

Elizabeth E. Rueschhoff  <http://orcid.org/0000-0002-8268-0905>

Viknesh Sivanathan  <http://orcid.org/0000-0001-7163-4543>

Sara S. Tolsma  <http://orcid.org/0000-0002-2687-4457>

Vassie C. Ware  <http://orcid.org/0000-0003-1604-5628>

Jackie M. Washington  <http://orcid.org/0000-0003-4287-8615>

Daniel Williams  <http://orcid.org/0000-0002-7900-480X>

Ellen M. Wisner  <http://orcid.org/0000-0003-0733-0021>

AUTHOR CONTRIBUTIONS

Tamarah Lynne Adair, Data curation, Formal analysis, Writing – review and editing | Swati Agrawal, Data curation, Formal analysis, Writing – review and editing | Yesmi P. Ahumada-Santos, Data curation, Formal analysis, Writing – review and editing | Regina Alvarez, Data curation, Formal analysis, Writing – review and editing | Kirk R. Anders, Data curation, Formal analysis, Writing – review and editing | Mauricio Antunes, Data curation, Formal analysis, Writing – review and editing | Mary Ayuk, Data curation, Formal analysis, Writing – review and editing | María Elena Báez-Flores, Data curation, Formal analysis, Writing – review and editing | Dondra Bailey, Data curation, Formal analysis, Writing – review and editing | Frederick N. Baliraine, Data curation, Formal analysis, Writing – review and editing | Mitchell F. Balish, Data curation, Formal analysis, Writing – review and editing | Christa Tobey Bancroft, Data curation, Formal analysis, Writing – review and editing | Michèle Barmoy, Data curation, Formal analysis, Writing – review and editing | Tonya C. Bates, Data curation, Formal analysis, Writing – review and editing | Andrea Beyer, Data curation, Formal analysis, Writing – review and editing | Suparna Bhalla, Data curation, Formal analysis, Writing – review and editing | Dave Bollivar, Data curation, Formal analysis, Writing – review and editing, Conceptualization, Writing – original draft | Lisa M. Bono, Data curation, Formal analysis, Writing – review and editing | Rebecca Bortz, Data curation, Formal analysis, Writing – review and editing | Tejas Bouklas, Data curation, Formal analysis, Writing – review and editing | Kristen Butela, Data curation, Formal analysis, Writing – review and editing | Christine A. Byrum, Data curation, Formal analysis, Writing – review and editing | Megan Carroll, Data curation, Formal analysis, Writing – review and editing | Steven Michael Caruso, Data curation, Formal analysis, Writing – review and editing | Nancy Castro, Data curation, Formal analysis, Writing – review and editing | Kari Clase, Data curation, Formal analysis, Writing – review and editing | Kristen Clermont, Data curation, Formal analysis, Writing – review and editing | Sean T. Coleman, Data curation, Formal analysis, Writing – review and editing | Pamela L. Connerly, Data curation, Formal analysis, Writing – review and editing | Amaya M. Garcia Costas, Data curation, Formal analysis, Writing – review and editing | Steven Gaines Cresawn, Data curation, Formal analysis, Writing – review and editing | Tom D'Elia, Data curation, Formal analysis, Writing – review and editing | Aimee Danley, Data curation, Formal analysis, Writing – review and editing | Megan K. Dennis, Data curation, Formal analysis, Writing – review and editing | Arturo Diaz, Data curation, Formal analysis, Writing – review and editing, Conceptualization, Writing – original draft | Erin L. Doyle, Data curation, Formal analysis, Writing – review and editing | Iain Duffy, Data curation, Formal analysis, Writing – review and editing | Dustin Cole Edwards, Data curation, Formal analysis, Writing – review and editing | Elvira Eivazova, Data curation, Formal analysis, Writing – review and editing | Eric Engstrom, Data curation, Formal analysis, Writing – review and editing | Christy Fillman, Data curation, Formal analysis, Writing – review and editing, Conceptualization, Writing – original draft | Christine L. Fleischacker, Data curation, Formal analysis, Writing – review and editing | Victoria J. Frost, Data curation, Formal analysis, Writing – review and editing | Julie Torruellas Garcia, Data curation, Formal analysis, Writing – review and editing | Bryan P. Gibb, Data curation, Formal analysis, Writing – review and editing | Alyssa Mae Gleichsner, Data curation, Formal analysis, Writing – review and editing | Sharon Gusky, Data curation, Formal analysis, Writing – review and editing | Rebekah F. Hare, Data curation, Formal analysis, Writing – review and editing | Danielle Heller, Data curation, Formal analysis, Writing – review and editing | Lee E. Hughes, Data curation, Formal analysis, Writing –

review and editing | Jacob D. Kagey, Data curation, Formal analysis, Writing – review and editing | Bridgette Kirkpatrick, Data curation, Formal analysis, Writing – review and editing | Karen K. Klyczek, Data curation, Formal analysis, Writing – review and editing | Kathryn P. Kohl, Data curation, Formal analysis, Writing – review and editing | Hari Kotturi, Data curation, Formal analysis, Writing – review and editing | Julia Y. Lee-Soety, Data curation, Formal analysis, Writing – review and editing, Conceptualization, Writing – original draft | Heather Lindberg, Data curation, Formal analysis, Writing – review and editing | Bhaswati Manish, Data curation, Formal analysis, Writing – review and editing | Sergei A. Markov, Data curation, Formal analysis, Writing – review and editing | Matthew David Mastropaolo, Data curation, Formal analysis, Writing – review and editing | James T. Melton, Data curation, Formal analysis, Writing – review and editing | Jon Mitchell, Data curation, Formal analysis, Writing – review and editing | Sally Dixon Molloy, Data curation, Formal analysis, Writing – review and editing | Denise L. Monti, Data curation, Formal analysis, Writing – review and editing, Conceptualization, Writing – original draft | Tiara Perez Morales, Data curation, Formal analysis, Writing – review and editing | María Alejandra Mussi, Data curation, Formal analysis, Writing – review and editing | Holly Nance, Data curation, Formal analysis, Writing – review and editing | Fernando Nieto, Data curation, Formal analysis, Writing – review and editing | Imade Yolanda Nsa, Data curation, Formal analysis, Writing – review and editing | Shallee T. Page, Data curation, Formal analysis, Writing – review and editing | Ricardo Parra-Unda, Data curation, Formal analysis, Writing – review and editing | Vipaporn Phuntumart, Data curation, Formal analysis, Writing – review and editing | Brett E. Pickett, Data curation, Formal analysis, Writing – review and editing | Richard Scott Pollenz, Data curation, Writing – original draft, Writing – review and editing, Conceptualization, Formal analysis | Sarah N. Reardon, Data curation, Writing – review and editing, Formal analysis | Jessica Rocheleau, Data curation, Formal analysis, Writing – review and editing | Ombeline Rossier, Data curation, Writing – review and editing, Formal analysis | Naomi Smith Rowland, Data curation, Writing – review and editing, Formal analysis | Adam D. Rudner, Data curation, Formal analysis, Writing – review and editing | Elizabeth E. Rueschhoff, Data curation, Formal analysis, Writing – review and editing | Viknesh Sivanathan, Conceptualization, Data curation, Formal analysis, Writing – original draft, Writing – review and editing | Martha Smith Caldas, Data curation, Formal analysis, Writing – review and editing | C. Nicole Sunnen, Data curation, Formal analysis, Writing – review and editing, Conceptualization, Writing – original draft | Sarah J. Swerdlow, Data curation, Formal analysis, Writing – review and editing | Kissaou Tchedre, Data curation, Formal analysis, Writing – review and editing | Kara Thoemke, Data curation, Formal analysis, Writing – review and editing | Deborah Tobiason, Data curation, Formal analysis, Writing – review and editing | Sara S. Tolsma, Data curation, Formal analysis, Writing – review and editing | Tara Turley-Stoulig, Data curation, Formal analysis, Writing – review and editing | Quinn Vega, Data curation, Formal analysis, Writing – review and editing, Conceptualization, Writing – original draft | Patricia Waikel, Data curation, Formal analysis, Writing – review and editing | Catherine Ward, Data curation, Formal analysis, Writing – review and editing | Vassie C. Ware, Data curation, Formal analysis, Writing – review and editing | Jackie M. Washington, Data curation, Formal analysis, Writing – review and editing | Daniel Eric Westholm, Data curation, Formal analysis, Writing – review and editing | Beth Wilkes, Data curation, Formal analysis, Writing – review and editing | Daniel Williams, Data curation, Formal analysis, Writing – review and editing | Elizabeth Williams, Data curation, Formal analysis, Writing – review and editing | Ellen M. Wisner, Data curation, Formal analysis, Writing – review and editing

ADDITIONAL FILES

The following material is available [online](#).

Supplemental Material

Appendix 1 (jmbe00313-25-S0001.docx). Designing an accessible scientific poster rubric.

Appendix 2 (jmbe00313-25-S0002.docx). Writing a genome announcement rubric.

Appendix 3 (jmbe00313-25-S0003.docx). Poster checklist.

Appendix 4 (jmbe00313-25-S0004.xlsx). Data collection template.

Appendix 5 (jmbe00313-25-S0005.docx). Manuscript template.

Appendix 6 (jmbe00313-25-S0006.docx). Genome announcement checklist.

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